

KIMS Insurance Packages — Excel vs DB Reconciliation Report

Date: 2026-07-13 **Source of truth:** /Users/apple/Downloads/inputs/KIMS Insurance Packages .xlsx **Compared against:** deployed DEV engine (fc-estimate-dev.figitalabs.com), GET /api/packages/detail per package (backed by fc.v_package_runtime_lookup — package_amount + room_rates_jsonb) **Scope:** ALL 642 Excel rows were looked up in the DB by service code (not a sample).

1. Executive summary — "how many times is it wrong"

Metric	Value
Excel package rows (6 insurer sheets)	642
Found in DB (by exact service code + tariff)	484 (75.4%)
Absent from DB entirely	158 (24.6%) — incl. the entire HDFC sheet (105 rows)
Packages where all room-tier rates match Excel	156 (24.3% of Excel; 32.2% of those found)
Room-tier price cells compared (Gen/Twin/Single)	1,452
Tier cells wrong or missing in DB	980 (67.5%) — 920 wrong amounts + 60 missing tiers

Only Bajaj is fully correct. Star is ~89% correct. GIPSA, Medi Assist and ICICI are wrong on essentially every package, and HDFC does not exist in the DB at all.

The premise is *partially* outdated in one respect: the DB **does** already store per-room-type rates (room_rates_jsonb , 3–4 tiers per package) alongside the single package_amount . The problem is that (a) package_amount is what the estimate engine surfaces and it is the **General Ward** rate (or, for GIPSA, an unexplained higher figure), and (b) for 3 of 6 insurers the stored tier rates come from an **older tariff revision** than the Excel.

2. Excel structure

Workbook has **6 sheets, one per insurer/TPA**. Every sheet is per-room-tier — confirming your instruction that tariffs differ by room type and provider.

Sheet	Rows	Code column	Room-tier columns	Other columns
GIPSA	210	SERVCODE	GW / TWIN / SINGLE	PACKAGE NAME, LOS, EXCLUSIONS
HDFC	105	EQUIVALENT CD	GENERAL / TWIN / SINGLE	PACKAGE NAME, LOS, Exclusions
MEDI ASSIST	84	SERVICE CODE	General ward / Semi Private / Private	PACKAGE NAME, LOS, Exclusion
STAR	120	EQUSSERVICECD (x2, duplicated header)	GENERAL / TWIN / SINGLE	Tariff name = STAR_HYD_PKG_23 , TARIFFCD = TR176 , SERVICENAME, EXCLUSION
ICICI	71	EQUIVALENT CD	GENERAL / TWIN / SINGLE / ICCU / DELUXE / SUITE	Service name, LOS, Exclusions
BAJAJ	52	SERVICE CODE	GENERAL / TWIN / SINGLE / ICCU / DELUXE / SUITE	SERVICENAME, LOS, Exclusion

Semantics observed:

- **Service codes are KIMS HIS codes** (e.g. CAR0122 , CTH0059) and match DB package_code directly — this made exact-code matching possible for all rows.
- **GIPSA sheet** is the GIPSA PSU-insurer rate card (structured multipliers: TWIN = GW × 1.125, SINGLE = GW × 1.25 on almost every row; flat/day-care packages 1.0×).
- **ICICI upper tiers are formulaic**: ICCU = SINGLE, DELUXE = SINGLE × 1.2, SUITE = SINGLE × 1.35 on **all 71 rows**.

- **BAJAJ upper tiers are capped:** ICCU = DELUXE = SUITE = SINGLE on all 52 rows (rate stops escalating above Single).
- **LOS** is mostly numeric days; some rows say `day care` / `DAYCARE` (free text).
- GIPSA has two rows with malformed codes containing spaces (`E N T0017` , `E N T0018`) — cleaned to `ENT0017/18` for lookup.
- Duplicate codes for the same procedure are common (e.g. three CAG codes `CAR0122/CAR5153/CAR5158` at the same price).

3. Match methodology

1. Parsed all 6 sheets with openpyxl (642 data rows).
2. Mapped each sheet to the DB tariff via `GET /api/lookup/organizations` :

Excel sheet	DB tariff	DB tariff name	Organization used
GIPSA	TR290	KIMI_GIPS_24(C)	ORG56 New India Assurance (same card serves National/Oriental/United)
HDFC	TR286	HDFC_HYD_23(C)	ORG126 HDFC Ergo
MEDI ASSIST	TR285	MEDI ASST-23(C)	ORG1197 Aditya Birla (same card: Go Digit, Manipal Cigna, MD India, Health Assist)
STAR	TR287	STAR_HYD_23(C)	ORG61 Star Health
ICICI	TR201	SSG_ICICI	ORG59 ICICI Lombard
BAJAJ	TR289	BAJA_HYD_23(C)	ORG57 Bajaj General

3. Called `/api/packages/detail?tariff_code&package_code&organization_cd` for **every** Excel row (642 calls; 54 transient 502s retried successfully). Note: `organization_cd` is **required** for insurance tariffs — without it the lookup view only matches org-blank (cash) rows and returns `no_package_exists` .
4. Compared Excel GW/Twin/Single against DB `room_rates_jsonb` tiers and against the single `package_amount` .

DB room-category codes per tariff (they are not uniform):

- GIPSA / STAR / MEDI ASSIST: `multi_sharing_general_ward` , `twin_sharing_ac_single_room_non_ac` , `deluxe_single_room_ac_private` (Medi Assist additionally has `general_ward_add_on` / `twin_sharing_add_on` / `single_room_add_on` per-day add-on amounts of 7,000/9,000/10,000)
- ICICI (TR201): `general` , `triple` , `twin` , `single_ac` (4 tiers)

- BAJAJ: `general` , `twin` , `single_deluxe`

Coverage: 484/642 Excel rows (75.4%) resolved to a DB package with amounts. This is a census of the Excel, not a sample; the only untested combinations are DB packages *not* present in the Excel (312 GIPSA, 176 Star etc. in DB vs 210/120 in Excel — the Excel is a subset).

4. Results per insurer

4.1 BAJAJ (TR289) — fully correct

52/52 found; all General/Twin/Single tier rates match the Excel exactly.

`package_amount` = General tier. Excel's ICCU/DELUXE/SUITE columns are identical to SINGLE, so the DB's 3-tier model loses nothing here.

4.2 STAR (TR287) — ~89% correct, 13 packages wrong

116/120 found (missing: `NES5099` ant. cervical discectomy, `NES5028` post. cervical discectomy; 2 others resolved on retry). 103 packages match on all three tiers. **13 packages mismatch (35 wrong tier cells)** — pattern: `package_amount` was updated to the Excel figure but `room_rates_jsonb` still holds the old revision:

Code	Package	Excel G/T/S	DB tiers G/T/S	DB package_amount
NES5208	Laminectomy & discectomy — cervical	225,225 / 251,213 / 277,200	150,150 / 167,475 / 184,800	225,225 (= Excel GW)
NES5209	Laminectomy & discectomy — dorsal	same	same stale	225,225
ORT0052	THR bilateral	253,047 / 281,165 / 312,933	168,698 / 187,443 / 208,622	253,047
ORT5531	TKR bilateral	same	same stale	253,047
URO5138	PCNL bilateral	174,875 / 193,283 / 211,691	116,583 / 128,855 / 141,127	174,875
URO5116	RIRS unilateral	72,600 / 79,200 / 85,800	60,500 / 66,000 / 71,500	72,600
URO5132 / URO5079	Orchiectomy / orchiopexy bilateral	66,000 / 74,250 / 82,500	52,800 / 59,400 / 66,000	66,000
SGA5085 / SGA5086	Hydrocelectomy bilateral / unilateral	66,000... / 44,000...	44,000... / 52,800...	79,200 / 52,800 — rates look swapped between uni- and bilateral
GYN0110	Vaginal hysterectomy	116,424 / 127,512 / 138,600	113,600 / 127,800 / 142,000	116,424
ORT5205	DHS / hemiarthroplasty	Twin 112,091	Twin 112,901	— digit transposition (091 vs 901) in one source
URO0023	Cystoscopy + DJ stent removal	Twin 22,000	Twin 22,500	20,000

4.3 GIPSA (TR290) — 0 packages fully correct

201/210 found (9 absent, incl. Normal Delivery GYN0023 , Cataract-Phaco OPT0260/0262 , PCNL UR00187/UR05372 , prostate Holmium UR05183 , SGA5194).

- **Every found package mismatches.** DB room_rates_jsonb holds a much older/lower rate set (e.g. CABG CTH0059 : Excel 225,600/253,900/282,100 vs DB tiers 162,900/179,200/195,500).
- The two sources even use **different escalation multipliers**: Excel Twin/Single = GW × 1.125 / × 1.25; DB tiers = base × 1.1 / × 1.2.
- **Systematic ratio found:** for 188/201 packages, **Excel GW = DB package_amount × 0.90 exactly** (rounded to ₹100). i.e. the DB single amount is ~11.1% above the Excel GW rate — either the DB holds an undiscounted 2024 rate (tariff name is KIMI_GIPS_24) and the Excel applies a 10% GIPSA discount, or the Excel is the negotiated card and the DB is inflated. 9 packages match at 1.0; 4 are irregular.
- 20 found packages have **no room tiers at all** in the DB (60 missing cells).
- Largest deltas (Excel GW vs DB GW tier): Renal transplant UR05365/5413 Excel 703,800 vs DB 385,000 (**Δ +318,800**); small-bowel resection SGA5379 265,900 vs 46,200; craniotomy NES5110/5111 239,700 vs 63,800; decompressive craniectomy NES0073 247,300 vs 77,000.

4.4 MEDI ASSIST (TR285) — 1 package fully correct

79/84 found with amounts (5 absent, incl. 2 CAG variants and nasal bone reduction). **78/79 mismatch.**

- **Systematic ratio:** Excel = DB tier × **1.18** on 222/234 tier cells (mean 1.19). The Excel is plainly a newer revision of the same card with a uniform **+18% uplift** the DB never received.
- DB uniquely carries per-day room **add-on** tiers (7k/9k/10k) with no Excel counterpart.
- **package_amount** = DB General tier (so doubly stale).

4.5 ICICI (TR201 SSG_ICICI) — 0 correct, likely the wrong contract entirely

Only 36/71 found (**35 absent = 49%**), and all 36 found mismatch on every tier with **scattered ratios (1.3×–3.2×)** — no uniform uplift. DB tier structure also differs (general/triple/twin/single_ac vs Excel's 6 columns). Conclusion: the DB's SSG_ICICI tariff is a **different (older/corporate SSG) agreement**, not the ICICI Lombard card in the Excel. Examples: AVR/MVR Excel GW 304,800 vs DB 169,000; CABG 294,000 vs 169,000; bilateral hip 261,000 vs 130,000.

4.6 HDFC (TR286) — completely absent

Tariff TR286 `HDFC_HYD_23(C)` exists and is linked to ORG126 HDFC Ergo, but has **0 packages** in the DB. All 105 Excel rows returned `no_package_exists`. The whole HDFC card needs loading.

5. Aggregates — your numbers

Insurer	Excel rows	In DB	Absent	Fully matching	Wrong (found but ≠)	Wrong-cell rate
BAJAJ	52	52	0	52 (100%)	0	0 / 156
STAR	120	116	4	103 (86%)	13	35 / 348 (10%)
GIPSA	210	201	9	0	201	603 / 603 (100%)
MEDI ASSIST	84	79	5	1	78	234 / 237 (99%)
ICICI	71	36	35	0	36	108 / 108 (100%)
HDFC	105	0	105	0	—	—
Total	642	484 (75%)	158 (25%)	156 (24%)	328 (51%)	980 / 1,452 (67.5%)

Systematic patterns (root causes, not random noise):

1. **HDFC card never loaded** (105 rows).
2. **GIPSA tier rates are an old revision** with old multipliers (1.1/1.2 vs 1.125/1.25), while `package_amount` sits ~11% *above* the Excel GW (Excel = amount × 0.90 on 94% of rows).
3. **Medi Assist is uniformly one revision behind** (Excel = DB × 1.18).
4. **DB "ICICI" is a different contract** (SSG_ICICI) — half the Excel's packages missing, prices unrelated.
5. **Star: `package_amount` refreshed, `room_rates_jsonb` not** on 13 packages — the two fields are drifting independently.
6. **Engine surfaces one number** — `package_amount`, which (where determinable) equals the **General Ward** tier — so any twin/single patient is under-quoted by 10–35%

even where the tiers are stored correctly.

7. Readiness is a separate axis: of the 484 found, 175 are `not_ready` (blockers = missing FC-history/template mapping), independent of price correctness.

6. Schema recommendation — per-room package amounts

The plumbing half-exists (`room_rates_jsonb` on `fc.package_master` → `fc.v_package_runtime_lookup`). Recommended target state:

1. **Normalise room rates into a child table** instead of (or materialising) the JSONB:

sql

```
CREATE TABLE fc.package_room_rate (
  tariff_code      text NOT NULL,
  package_code     text NOT NULL,
  room_category_code text NOT NULL, -- canonical enum below
  amount           numeric(12,2) NOT NULL,
  rate_basis       text,             -- 'contract' | 'derived_multiplier'
  effective_from   date NOT NULL,
  effective_to     date,             -- versioning: never overwrite,
close old row
  source_ref       text,             -- e.g. 'KIMS Insurance
Packages.xlsx / STAR / row 14'
  PRIMARY KEY (tariff_code, package_code, room_category_code,
effective_from)
);
```

2. **Canonical room-category enum** covering every tier seen in either source: `general_ward`, `semi_private_twin`, `private_single`, `iccu`, `deluxe`, `suite` — plus a per-tariff mapping table (`fc.tariff_room_category_map`) because each insurer names/collapses tiers differently (Bajaj caps everything above Single; ICICI derives Deluxe/Suite as 1.2x/1.35x Single; Medi Assist calls them GW/Semi-Private/Private).
3. **Per-day room add-ons** as a rate-type on the same table (Medi Assist's 7k/9k/10k) rather than a bespoke column.
4. **Deprecate scalar `package_amount` as the quoted figure.** Keep it only as a fallback and define it explicitly (= `general_ward` rate). The coverage engine should resolve (`tariff`, `package`, `patient's room_category`, `admission_date`) → rate, using the policy's room-cap logic (already a concept in the estimate flow) to pick the tier.
5. **Version, don't overwrite** — the Star drift (amount updated, tiers stale) and the GIPSA/Medi Assist "old revision" problems are exactly what `effective_from/to` + `source_ref` prevent.

6. ETL: this Excel can seed the table directly — codes align with `package_code` for 75% of rows out of the box.
-

7. Open questions for you

1. **GIPSA 0.9 factor** — Excel $GW = DB \text{ package_amount} \times 0.90$ on 188/201 packages. Is the Excel a 10%-discounted GIPSA card (and the DB's 2024 amount right but undiscounted), or is the Excel the signed rate and the DB inflated? Which multipliers are contractual — 1.125/1.25 (Excel) or 1.1/1.2 (DB)?
 2. **ICICI** — the DB only has `SSG_ICICI` (TR201). Is the Excel ICICI sheet a *new* ICICI Lombard agreement that should become a new tariff, or a replacement for TR201?
 3. **STAR tariff identity** — Excel says `TARIFFCD TR176 / STAR_HYD_PKG_23`; the DB serves Star from `TR287 STAR_HYD_23(C)`. Same agreement, different code — which is canonical? (Prices agree 89%, so probably the same card.)
 4. **Star hydrocelectomy** `SGA5085 / SGA5086` — unilateral/bilateral rates appear swapped between Excel and DB; which is right? Also `ORT5205` Twin 112,091 vs 112,901 (digit transposition — which source has the typo?) and `UR00023` Twin 22,000 vs 22,500.
 5. **HDFC** — confirm the sheet is current so we can load all 105 packages into TR286.
 6. **ICICI/Bajaj upper tiers** — should ICCU/Deluxe/Suite be stored explicitly, or derived (ICICI: $1.0/1.2/1.35 \times \text{Single}$; Bajaj: capped at Single)?
 7. **Medi Assist +18%** — is the Excel the currently effective revision (effective when?), and do the DB's per-day room add-ons (7k/9k/10k) still apply on top?
 8. **Missing packages** (9 GIPSA, 4 Star, 5 Medi Assist, 35 ICICI listed in section 4) — add to the catalogue, or intentionally dropped from the contracts?
 9. The Excel's `L0S` includes free text (`day care`) — confirm day-care handling for pre/post-day logic.
-

8. Honesty about limits

- Matching was by **exact service code** within the mapped tariff — stronger than name matching, but if a package was re-coded between revisions it would show as "absent" rather than "renamed" (likely contributes to some of the 158 absences, esp. ICICI).
- Only one organization per tariff was queried (e.g. New India for GIPSA); other orgs on the same tariff are assumed to share the card, as the org lookup indicates.
- DB values came via the runtime view through the API, not raw SQL — anything the view filters out (inactive rows, other room-category codes) is invisible here.
- "±" thresholds: amounts compared exactly ($< ₹0.50$); the 0.90 / 1.18 ratios allow $\pm 1\%$ for the ₹100 rounding visible in the Excel.
- The DB contains packages *not* in the Excel (e.g. 312 GIPSA in DB vs 210 in Excel); those were not audited — this report only answers "is what the Excel says reflected in

the DB".